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Review Article

Review of the studies on emission evaluation approaches for operating vehicles

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HIGHLIGHTS

- Reviewed the vehicle emission measurements from laboratory, on-road, and tunnel studies.
- Summarized the factors affecting the emission evaluation of operating vehicles.
- Analyzed the applications of the emission evaluation models of the vehicles operating at road segments and intersections.
- Proposed further research for the emission evaluation approaches of operating vehicles.

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ABSTRACT

With the increasing international consensus concerning the negative effects of climate change, reducing greenhouse gases has become a higher priority in government policies and research committees. The transportation sector generates approximately 29% of the total greenhouse gas emissions and 25% of the global energy related carbon dioxide (CO₂) emissions. Thus, it is essential to understand the influencing factors of vehicle emissions and establish a corresponding evaluation model for the emission estimation of operating vehicles. This paper reviews different methods of measuring vehicle emissions, including laboratory measurements, on-road measurements, and tunnel measurements. Then, we summarize the factors affecting the emission evaluation of operating vehicles based on the vehicle operating features and road environment. Finally, the applications of vehicle emission evaluation models are analyzed, including the emission assessment of the vehicles operating at road segments and intersections. Based on this review, one can conclude that selecting different measurements will significantly impact the assessment of the vehicle emission results and the applicable scope of the measurements. Considering the different influencing factors of the operating vehicle emissions will have an impact on the model application of the vehicle emission evaluation. Moreover, several analysis methods for new technology vehicles can compensate for the potential lack in connectedness with the rapid development of new energy vehicle technology and the improvement of intelligent transportation systems (ITS).

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1. Introduction

The fifth assessment report of the intergovernmental panel on climate change (IPCC) indicates that the global surface temperature has increased by 0.89 °C in the past 100 years (Alexander et al., 2013). More than 90% of the global warming in the past 50 years is related to increases in greenhouse gases from fuels used by humans. In 2015, the concentrations of greenhouse gases, such as CO₂, methane (CH₄), and nitrous oxide (N₂O) in the atmosphere increased by 44%, 156%, and 21%, respectively, compared with these concentrations in 1750 (Alexander et al., 2013; EPA, 2018).

The natural greenhouse gas is primarily a function of the concentration of water vapor, CO₂, CH₄, N₂O, and ozone (O₃). Human emitting greenhouse gases include water vapor, CO₂, CH₄, N₂O, and other trace gases in the atmosphere. Besides, several other substances including carbon monoxide (CO), nitrogen dioxide (NO₂), sulfur dioxide (SO₂), volatile organic compounds (VOCs) and nitrogen oxides (NO_x) also have spatially variable radiative forcing effects on global radiation. Fig. 1 presents the relative contribution of the pollutants to total greenhouse gas emissions in 2019 in the US (EPA, 2018). The largest portion of greenhouse gas directly emitted by human activities was CO₂ accounting for nearly 80.1% of total greenhouse gases. CH₄, N₂O, and other trace gases represent approximately 10.1%, 7.0%, and 2.8%, respectively, of greenhouse gas emissions. Global carbon dioxide emissions are still increasing, as shown in Fig. 2, which has led to an increase in the global average surface temperature; the sea level has also risen at a faster rate (IEA, 2019). If the trend of global warming continues, there will be more natural disasters, which will affect the survival and development of humanity. In 2010, China became the world's largest energy consumer (Lyu et al., 2016) and was under pressure to reduce emissions to assist with global climate control.

According to commonly used economic sector categories to characterize greenhouse gas emissions, they are allocated

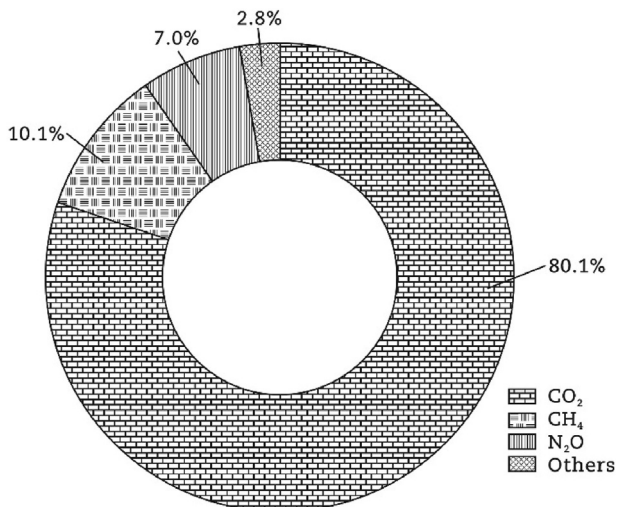


Fig. 1 – Pollutants as a portion of all greenhouse gas emissions in 2019, US.

to five economic sectors (EPA, 2018) which are transportation, electricity, industry, commercial and residential, and agriculture. Fig. 3 illustrates the greenhouse gas emissions by economic sectors in 2019 in the US. Obviously, the transportation sector generates the largest source of greenhouse gas emissions from human activities. Transportation is an essential industry for national economic and social development, and it is one of the fastest-growing fields in energy consumption and emissions in the world. The global-energy-related carbon dioxide emissions are rising yearly (Fig. 4) and nearly 25.0% of the world's energy-related CO₂ emissions are generated by transportation every year (Zhang et al., 2019). Road transport accounts for 74.0% of transportation CO₂ emissions (Gül et al., 2020). The transportation sector accounts for 25.0% of the total emissions, and road traffic accounts for 18.5% of the total CO₂ emissions. Aviation accounts for only 11.6% of the transportation emissions and about 2.9% of the total global emissions. The contribution of shipping was 10.6% of the transportation emissions. Rail travel and freight emissions are not significant, accounting for only 1.0% of the transportation emissions. Other types of transport, such as pipelines, accounted for 2.2% (Gül et al., 2020). Therefore, in terms of solving global climate problems and implementing emission reduction measures, ground transportation can be an important area of study for energy conservation and emission reduction in China.

In the past 20 years, the carbon dioxide emissions of different transport modes of road traffic have shown a noticeable growth trend (Gül et al., 2020). Among them, passenger cars and trucks have the most significant increase, as shown in Fig. 5. Therefore, a comprehensive analysis of the characteristics of road traffic emissions changes and effective methods of emission evaluation could fundamentally provide support for road transportation emission control. It could help achieve the goal of reducing emissions from the transportation industry, and so it is an important area to explore to limit peak carbon dioxide emissions and concentrations (Seshadri, 2015).

This study assesses the existing research on the emission evaluation approaches of vehicles operating at segments and intersections. It explores the characteristics and influencing factors of operating vehicle emissions, analyzes the

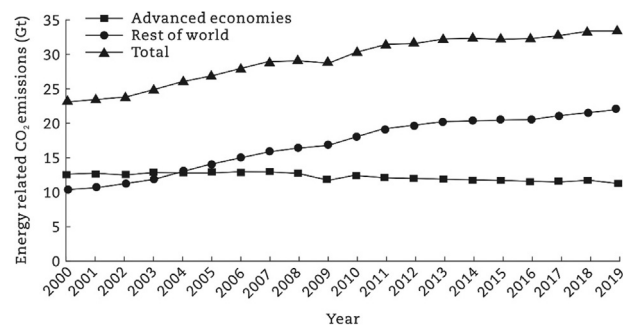


Fig. 2 – Global energy related carbon dioxide emissions, 2000–2019.

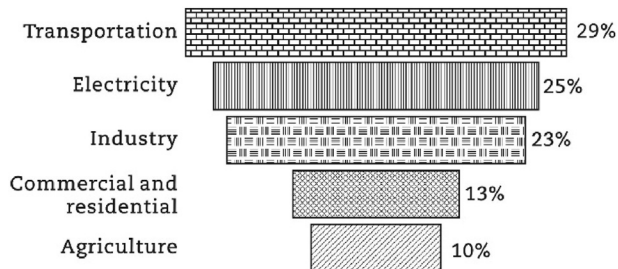


Fig. 3 – Greenhouse gas emissions by economic sector in 2019, US.

methods of emission evaluation of operating vehicles, and highlights potential directions of future research. In summary, we hope to address the following questions through this review article.

- (1) What are the advantages and limitations of various transportation carbon emission measurements?
- (2) Taking the vehicle and road into account as the operating subjects, what are the evaluation objects of the different types of influencing factors and what are the corresponding emission indicators of each influencing factor?
- (3) What are the characteristics and implementation objectives of each emission evaluation method? Based on those questions, the remainder of this review is structured as follows. Section 2 summarizes the emission evaluation models of each measurement, and compares the various vehicle emission evaluation methods. Section 3 presents different factors influencing the application of the operating vehicle emission evaluations. Section 4 presents the application of the emission evaluations for operating vehicles, which is helpful for the systematic analysis of the emissions of various operating vehicles and the realization of low-carbon transportation.

2. Vehicle emission measurements and models

Vehicle exhaust emission measurement methods are mainly divided into laboratory measurements and real-world measurements. Laboratory measurements are conducted for vehicle emission testing in a controlled environment, and the real-world measurements include the tunnel test, remote sensing test, near-road measurement, and on-board test (Smit and Kingston, 2019). These tests have different functions with their own advantages and can meet different research needs. Most emission models were developed for determining emission factors from emission measurement methods, such as tunnel studies, on-road measurements, and laboratory measurements (Smit et al., 2010).

2.1. Laboratory-measurement-based emission models

In the laboratory, exhaust emissions are measured under controlled conditions using vehicles capable of operating in a certain manner. The studied vehicles follow a specific driving pattern in which the instantaneous speed specifically follows the speed-time profile to simulate a chosen driving behavior for a specific vehicle type or a typical period of the day. Models built on lab measurements are widely used and serve as the primary foundation of current emission models (Fomunung et al., 2001; Joumard et al., 2003; Lacour et al., 2001). The dynamometer testing of a large number of laboratory vehicles has been employed to estimate the emission rates in terms of the instantaneous or average data based on multiple simulated driving behaviors. It can also be used to evaluate the impacts of various controlled driving behaviors and operating conditions on emissions. Moreover, it has been used for the emission standards for emission testing procedures, as well as for evaluating regional emission inventories by aggregating the vehicle data of activities and emissions (Smit et al., 2009).

Various driving cycles were performed to characterize driving patterns under different operating conditions in a

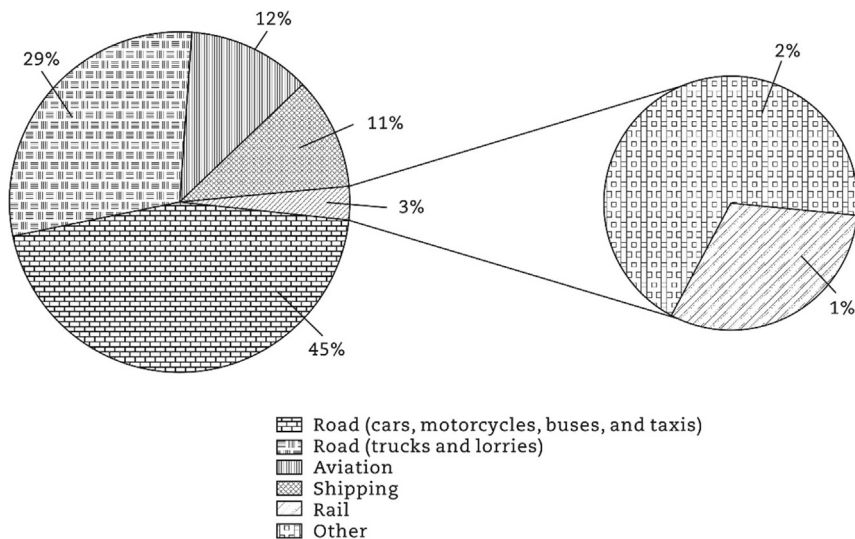


Fig. 4 – Global CO₂ emissions from transportation.

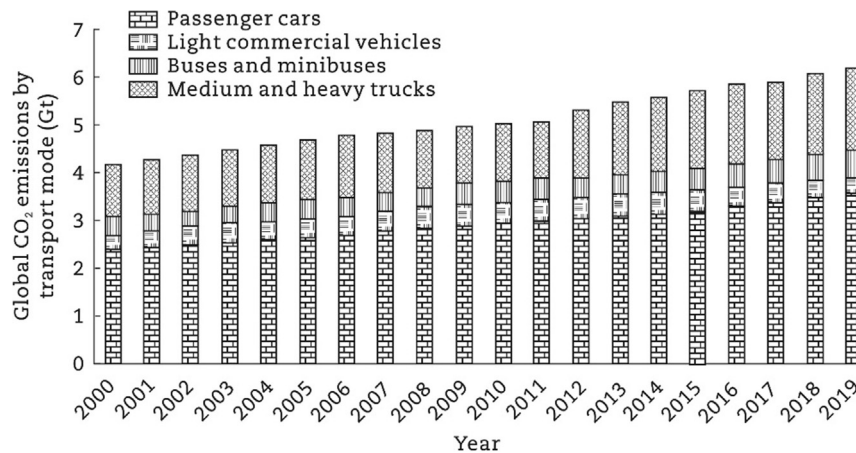


Fig. 5 – Global CO₂ emissions by transport mode, 2000–2019.

laboratory setting, and then these measurements were used to estimate the emission factors for new and in-use light-duty and heavy-duty fleets (Brzezinski et al., 1999). A laboratory dynamometer driving cycle test was carried out using a predetermined speed-time profile, which included the driving modes of a constant speed, accelerations, decelerations, starts, and stops under multiple driving conditions, and included the specific fleet, controlled fuel type, and temperature. An overall time-weighted average speed is typically used to present a driving cycle (Alpharetta et al., 2004). Some well-known models were developed based on the measurements mentioned earlier to evaluate emissions.

2.1.1. Averaged speed models

Mobile source factor model (MOBILE) and emission factor (EMFAC) are first-generation vehicle emission models developed by the US EPA and the California Air Resources Board (CARB).

The MOBILE model is a regulatory model used to calculate the average vehicle emissions. The model parameters are derived from the testing results of various in-use vehicles at different emission levels, as well as the emission results measured in the federal testing program (FTP), which is extensive and representative of the types of programs in the US. Using a large amount of measured data, the model can simulate the running conditions of local motor vehicles and background climate conditions to predict the actual emission factors. The system is relatively complete, and its database offers broad coverage, which has led to its wide range of use (Alpharetta et al., 2004).

EMFAC model can be used to predict regional inventories and projections for HC, CO, NO_x, CO₂, PM, SO_x, and fuel consumption. The driving-cycle-based emission factor method was used by the EMFAC modeling methodology for multiple modeling scales (Saharidis and Konstantzos, 2018). Vehicle emission rates were derived from the driving cycle tests that considered the vehicle type, pollutant type, fuel type, temperature, and humidity. A primary function of a vehicle's average travel speed was employed for the driving cycle

tests and sensor data. If a vehicle's average speed was different from that used in the standard drive cycle test, an adjustment factor (the speed correction factor (SCF), which was derived from a large number of driving cycle tests) was applied to adjust the emission rate based on various parameters, such as the vehicle type, vehicle year, pollutant type, and fuel type.

Another average speed model is computer programme to calculate the emissions from road transport (COPERT), which was developed by the European Commission (EC). COPERT estimates the emissions of air pollutants produced by different vehicle categories (passenger cars, light commercial vehicles, HDV, buses, motorcycles, and mopeds) (Denier van der Gon et al., 2013; Saharidis and Konstantzos, 2018). In COPERT 4, all major air pollutants and greenhouse gas emissions, including CO, NO_x, VOC, PM, NH₃, SO₂, CO₂, N₂O, and CH₄, emitted from vehicles can be estimated (Saharidis and Konstantzos, 2018). This model employs a significant amount of reliable experimental data, which are compatible with the statistical calibrations and parameter variables of different countries. COPERT uses the statistical data of fuel sales within a region as its basic data, which has better applicability in regions lacking basic traffic data. The COPERT model classifies vehicle types in detail and evaluates many kinds of pollutants. In addition, the interface friendliness of the model is better than that of the MOBILE model, which is more convenient for users to understand and operate (Cheng et al., 2011; Xu et al., 2009).

These models use a conceptually similar approach. They utilize statistical regression for the results of the FTP-based platform test, which considers the impact of the vehicle mileage, new vehicle technology level, deterioration factor, speed, air temperature, inspection and maintenance (I/M), and fuel quality emissions. However, these kinds of models only use the average speed to present the influence of the driving characteristics on emissions and use the speed correction factor to calculate the emission factors under non-FTP conditions. This approach produces weak operating characteristic results, which are important factors affecting

vehicle emissions. Moreover, the average speed emission models are designed for large regions or areas and need a large amount of data to be extrapolated beyond the initial collection. They are not capable of predicting microscopic emissions of second-by-second vehicle operation conditions with operating modes, including cruising, acceleration, deceleration, and idling (Barth et al., 2001; Frey et al., 2001), and instantaneous traffic characteristics, such as the dynamic traffic flow resulting from dynamically designed signal timing plans. Another limitation to the average speed modeling approach is the need to aggregate the vehicle's average speed and emissions; the model is not suitable for evaluating the emissions rate using the data with disaggregated properties from individual projects.

2.1.2. Instantaneous speed models

Some of the instantaneous speed emission models were developed to measure the correlation between instantaneous vehicle power and the emission factors to overcome the limitations of the average speed modeling approach. The comprehensive modal emissions model (CMEM) was developed by Barth et al. (2001) to evaluate the emission pollutants of CO, CO₂, HC, NO_x, and PM, as well as fuel consumption. In this model, the physical power demand method was employed to characterize the vehicle operating modes, which were then associated with various factors. The analytical formula used for predicting the second-by-second emissions and fuel consumption in different states of conditions utilized the properly functioning condition, deteriorated condition, and malfunctioning condition for multiple vehicle types and technology categories.

To predict the instantaneous fuel consumption and emissions for individual vehicles, Ahn et al. (2002) proposed a microscopic model of the instantaneous vehicle fuel consumption and emission rate named the Virginia Tech microscopic vehicle fuel consumption and emission model (VT-Micro). The model is used to calculate emissions of HC, CO, NO_x, and CO₂, and the fuel consumption with second-by-second vehicle characteristics. Based on the tests of the instantaneous vehicle speed, acceleration, emission rates, and fuel consumption for eight light-duty passenger cars and trucks, this model typically utilizes the instantaneous speed and acceleration to represent the characteristics of vehicle operation modes.

Two other well-known instantaneous speed emission models are related to the Artemis Project in Europe. The passenger car and heavy duty emission model (PHEM) was developed under the framework of the Artemis Project and Cost Action 346. The EMPA model was derived from the European Artemis and Particulates Projects and the handbook of emission factors for road transport (HBEFA) (Saharidis and Konstantzos, 2018). The PHEM model uses the parameters of driving resistances and the losses in the transmission system to determine the dynamics of the driving cycle and relies on the instantaneous engine power demand and engine speed to estimate the fuel consumption and emissions for various driving patterns (Rexeis et al., 2005). The EMPA model takes into account the size of the test vehicle's exhaust system and the measured modal data of the exhaust's volume flow (Boulter et al., 2007). Using the results from a large number of advanced tests, the test

Table 1 – Comparison of emission model features.

Model	Category	Main characteristic
MOBILE	Averaged speed model, developed by US EPA	Calculate the average vehicle emissions, parameters are derived from the in-use vehicles testing and FTP.
EMFAC	Averaged speed model, developed by US CARB	Predict regional inventories and projections for HC, CO, NO _x , CO ₂ , PM, SO _x , and fuel consumption.
COPERT 4	Averaged speed model, financed by EEA	All major air pollutants and greenhouse gas emissions, including CO, NO _x , VOC, PM, NH ₃ , SO ₂ , CO ₂ , N ₂ O, and CH ₄ , emitted from vehicles can be estimated; vehicle types include passenger cars, light commercial vehicles, HDV, buses, motorcycles, and mopeds.
CMEM	Instantaneous speed model, developed by NCHRP	Evaluate the emission pollutants of CO, CO ₂ , HC, NO _x , and PM, as well as fuel consumption.
VT-Micro	Instantaneous speed model, proposed by Ahn et al. (2002)	Calculate emissions of HC, CO, NO _x , and CO ₂ , and fuel consumption.
PHEM	Instantaneous speed model, originated from the Artemis Project	Estimate the fuel consumption and emissions for various driving patterns; determine the dynamics of the driving cycle and relies on the instantaneous engine power demand and engine speed.
EMPA	Instantaneous speed model, originated from the Artemis Project	Estimate the emissions of an individual vehicle in a short time, considering the vehicle's exhaust system size.
MODEM	Instantaneous speed model, originated from the Drive Project	Evaluate fuel consumption and CO, HC, NO _x and CO ₂ emissions for EURO I.
MOVES	Multi-scale model, developed by US EPA	Evaluate numerous pollutants for the US vehicle fleet of LDVs and a few HDVs.
IVE	Multi-scale model, developed by UCR	Estimate local air pollutants (criteria pollutants), greenhouse gas emissions, and toxic pollutants.

parameters and high-frequency measurement values were obtained to establish a model that can estimate the emissions of an individual vehicle in a short time.

The EMPA model does not include the instantaneous dynamic behavior of the engine when a vehicle operates in actual conditions, and this model was mainly derived from the LDV model (Saharidis and Konstantzos, 2018). Although the PHEM model fully considers the dynamic behavior of the engine, including the instantaneous engine speed and power (and it also includes the HDV map), it is used for modeling the engine's dynamic data based on the specified drive mode and fixed engine type, such as the Euro II and Euro III engines, and cannot accurately reflect the real-time state changes of engines in the real operation of various models (Rexeis et al., 2005). A comparison of model feature is illustrated in Table 1.

The modeling of emissions and consumption (MODEM) in urban areas was developed as an instantaneous emission model for EURO I vehicles to evaluate fuel consumption and CO, HC, NO_x and CO₂ emissions, with the limitation of the pollutant types and vehicle classes (Saharidis and Konstantzos, 2018).

2.1.3. Multi-scale models

The common multi-scale emission model motor vehicle emission simulator (MOVES) model was accepted by the federal and state governments as the official model for regulatory emission inventories estimation for the conformity analysis of transportation projects and air quality in the US emission factor data, which was derived from selected driving cycles represented by averaging vehicle activities and emissions on the basis of laboratory dynamometer testing for a large region, supported these macroscopic emission models.

The MOVES is a state-of-the-art vehicle emission factor model developed by the US EPA's office of transportation and air quality (OTAQ) based on a large number of dynamometer tests of FTP driving cycles and a large number of operating conditions, including fuel categories, vehicle types, vehicle age, temperature, humidity, and road grade. The vehicle specific power (VSP) method is employed as a core methodology in this model (EPA, 2010; Perugu, 2019). By considering various parameters impacting the fuel type, pollutant type, vehicle types, and vehicle age, the VSP determines the driving cycle bin, which directs itself to a specific driving cycle ID (EPA, 2009; Perugu, 2019). Then the emission factors are derived from the VSP binned driving cycle ID, which is related to a range of average speeds from the targeted vehicles for inventory analysis at the national and regional level, as well as emission estimations at the project level (Koupal et al., 2008). It can evaluate numerous pollutants, including GHG, VOCs, NO_x, PM, and CO. However, the limitation of this model's application is that the emission estimations are more applicable for the US's vehicle fleet of LDVs and a few HDVs (Saharidis and Konstantzos, 2018).

The University of California, Riverside (UCR), developed the international vehicle emissions (IVE) model. The core algorithm of the model is based on the MOBILE mode. By processing tens of thousands of sets of EPA, CARB, and UCR test

data, the basic emission factors based on the FTP conditions were obtained. By comparing the state element distribution of the target working conditions and FTP working conditions, the vehicle emission level under the target working condition (IVE, 2008) is obtained.

2.2. On-road measurement emission models

Driving-cycle-based models do not necessarily represent the operating mode of a vehicle and the correlating emissions on the road, which are related to improved operational impacts and dynamic progress. The driving cycle represented by the time-weighted average speed and averaged vehicle emissions is not suitable for analyzing the environmental impacts associated with dynamic traffic conditions, such as the conditions of the dynamic traffic flow at intersections with signal timing optimization. Thus, on-road measurement models for vehicle exhaust emission testing are a research hotspot (Bluett and Fisher, 2005; Elst et al., 2004; Guo et al., 2007a; Noland et al., 2006; Silva et al., 2006).

The on-road measurement analyzes exhaust emissions from real-world data using instantaneous real-time emission collection or the remote-sensing devices (RSDs) from large fleets of individual vehicles at certain locations passed by the vehicles in the road's network. This data can help to elucidate the correlation of the emissions and vehicles' behavior in the real world (Sjödén et al., 2017). Another advantage of applying the RSD is that it is possible to test the overall effectiveness and monitor the high exhaust emission vehicle in I/M programs.

However, the on-road measurement has not been extensively accepted as a method for calculating emissions for different types of vehicles or for different types of emissions, especially for heavy traffic across multiple lanes, because it limits the range of the operating conditions and controls the size and weight of the system. The relative costs for the measurement are another limitation. In prior research, this measurement was employed for the engine parameter test (Cicero-Fernández et al., 1997) or for estimating emission factors in a small system (Frey and Eichenberger, 1997).

To overcome the limitations of a prohibitively expensive approach to testing, the EPA developed a lower-cost model capable of evaluating vehicle behavior and emissions in a large testing system for light and heavy-duty vehicles (Fritz et al., 2001). Vojtisek-Lom and Cobb (1997) developed a portable, cheap, and non-invasive approach capable of measuring both vehicle activity and emissions. They employed a comparatively low-cost model to quantify gasoline and natural gas-fueled vehicle emissions, as well as the emissions of the shuttle buses at the University of Pittsburgh.

A modal emissions model predicting the fuel consumption and second-by-second vehicle emissions for light-duty vehicles was developed by the Center for Environmental Research and Technology at the University of California Riverside (UCR-CERT) for multiple vehicle types under different states of conditions (Barth et al., 2001). The final model was integrated into TRANSIMS based on a detailed simulation of household

Table 2 – Categorization of the factors influencing operating vehicle emissions.

Affecting factor	Second-level factor	Specific factor	Reference
Vehicular operating factor	Fuel factor	Fuel type	Eggleston et al. (2006); Li et al. (2014); Liu et al. (2016); Zhang et al. (2019)
		Fuel consumption	
	Speed factor	Travel distance	
		Engine type	
Road environmental factor	Segment factor	Average speed	Ajtay et al. (2005); Boulter et al. (2007); Correia et al. (2014); Jimenez-Palacios (1998); Saharidis and Konstantzos (2018); Sun et al. (2015); Ubiergo and Jin (2016)
		Instantaneous speed	
		VSP	
		Road elevation	
	Intersection factor	Road grade	Costagliola et al. (2018); Della Ragione and Giovanni (2016); Frey et al. (2008); Gallus et al. (2017); Prati et al. (2014), 2015; Wang et al. (2015); Wood et al. (2014)
		Tire-pavement	
		Signal timing	
		Signal coordination	
		Average delay	Benson (1984); Coelho et al. (2005); Garber and Hoel (2014); Hallmark et al. (2000); Huang et al. (2014); Liao (2013); Midenet et al. (2004); Pandian et al. (2009); Roupail et al. (2001); Zhang et al. (2014); Zhao et al. (2017)

and residential trips and vehicle operations (Willians et al., 1998) for the transportation system of an urban area.

In recent years, with the improvement of the instrument technology, on-board test system products that can be used in vehicle emission research have been greatly enriched.

For example, the EPA developed ROVER on-road emission monitoring to measure and evaluate the vehicle emission rate and distribution. The portable emission measurement system (PEMS) was developed by CATI to measure second-by-second mass emissions. SEMTECH products were developed via the partnership of Sensor and Ford (Hong et al., 2006).

With the deepening of vehicle emission research, more researchers are focusing on the instantaneous emission characteristics of vehicles on actual roads. Vehicle emission testing systems are being used in research in China, as well. Chen et al. (2005) used the SEMTECH-DS vehicle emission tester to carry out emission tests on heavy diesel vehicles on roads. Hui et al. (2007) applied RSD to evaluate the IVE model. On-board exhaust detection equipment (OEM-2100) was used to carry out the real road emission testing on the gas of heavy diesel vehicles in Beijing (Chen and Yu, 2007; Guo et al., 2007b).

2.3. Tunnel study measurements of vehicle emission models

Tunnel study measurements are used to measure the emissions from a large vehicle sample in relatively limited conditions, like high-speed driving not suitable for urban operating conditions and signalized arterials.

In some tunnel experiments, the concentration of pollutants was monitored inside and outside the tunnel, and the average emission factor level of the corresponding fleet was obtained according to the traffic flow parameters on site. Many tunnel experimental studies have been carried out in China (Huang et al., 2014; Liu et al., 2015; Wang et al., 2001). The tunnel experiment is easy to execute, and it is representative of the actual conditions to some extent because of the large number of vehicle samples involved in the test. Because of its low precision requirements, it can be used to obtain the average emission factor of the fleet. The main disadvantage is that the results are influenced by the background concentration of emissions, and it only

measures the conditions of certain vehicle types. Thus, it is often used for model validation under controlled conditions (Barlow and Boulter, 2009; Brousse et al., 2005; Colberg et al., 2005; John et al., 1999; Kirchstetter et al., 1996; Pierson et al., 1996; Sturm et al., 2001). Moreover, the tunnel measurements are one of the validation methods for the PHEM (Boulter et al., 2007).

3. Influencing factors of operating vehicle emissions

Operating vehicle emissions are the results of multi-factor interactions. Research on these vehicular emission factors varies, but the influencing factors of operating vehicle emissions in the literature differed significantly according to the research content. The data are abundant and include values for the fuel type, fuel consumption, travel distance, speed, road grade, road elevation, operating acceleration, restraint, and so on. In this paper, the various factors were categorized into the vehicle operating influencing factors and road environmental influencing factors (Table 2). The purpose of this classification is to illustrate the model application focus corresponding to the various influencing factors.

The vehicle operating factors mainly refer to the process of vehicle use considering some vehicle operation characteristics, such as fuel factors and speed factors. The main focus of an emission model for the road environmental factors is the vehicle's external environment during operation. Since the various indicators of a running vehicle interact with its external environment, the road indicator is an important external factor that can significantly affect the results of vehicle emissions (Frey et al., 2008). The road environmental factors were analyzed from the perspective of the road characteristics, including the road types, such as segments and intersections.

3.1. Vehicular operating factors

Vehicle operating factors are mainly divided into fuel-related factors and speed-related factors.

3.1.1. Fuel-related factors

Fuel-related factors are dominated by vehicle characteristics and taken into account the operational requirements during the vehicle's operation, including the fuel type, fuel consumption, travel distance, and engine type. Fuel-related factors include the different engine types, fuel types used, and fuel consumption during operation from the perspective of a running vehicle, and these factors directly impact the emissions of a certain travel distance. Theoretically, the engine type is the basis of a vehicle's emissions, and a larger engine will consume more fuel and produce more emissions than a smaller one. The engine type is also related to the fuel efficiency. The operating resistance of different engine types and the vehicle inertia at low speeds will have different effects on the fuel efficiency and emission factors (Li et al., 2014; Liu et al., 2016). In addition, fuel type is also an important indicator of emissions (Zhang et al., 2019). Different types of energy have varying carbon emission coefficients. At present, vehicles on the market are divided into traditional fossil fuel vehicles, alternative fuel vehicles, hybrid electric vehicles, pure electric vehicles, and fuel cell vehicles (Sharma et al., 2020; Zhang et al., 2019). Fuel consumption is a key factor in the vehicle emission model. It is an indicator revealing the characteristics of energy use during vehicle operation, and forms the basis of the emission factor. It is closely related to factors such as the travel mode and running distance.

Therefore, the modeling methods for calculating the carbon emissions given in the IPCC guidelines were based on the fuel consumption and comprehensively considered the vehicle type, fuel type, travel distance, and other factors. The main approaches to carbon emissions can be divided into the top-down and bottom-up models (Zhang et al., 2019).

The top-down model is described in Eq. (1) (Zhang et al., 2019).

$$E = \sum_i EF_i V_i \quad (1)$$

where E is the traffic carbon emissions and i is the fuel type in the top-down model, EF_i is the carbon emission factor, and V_i is the fuel consumption.

The bottom-up model can be used to calculate the emissions for different travel modes and vehicle types, as described by Eq. (2) (Zhang et al., 2019).

$$T = \sum_{ij} V_{ij} D_{ij} C_{ij} F_{ij} \quad (2)$$

where T is the traffic carbon emissions and i is the vehicle type in the bottom-up model, j is the energy source, V_{ij} is the number of vehicle i using the energy source j , D_{ij} is the travel distance of vehicle i using the energy source j in a certain time period, C_{ij} is the average energy consumption of vehicle i using the energy source j , and F_{ij} is the carbon emission factor of vehicle i using the energy source j .

3.1.2. Speed-related factors

Vehicle running speed is the basic index of emissions in many models, and it mainly includes the average speed and instantaneous speed. In the emission model based on the average speed or adjusted average speed, the determination of emissions is theoretically based on the emission coefficient

and average speed, while the average operating speed of a vehicle type can determine the average emission coefficient of a certain pollutant (Saharidis and Konstantzos, 2018). In addition, the simulation and parameter modifications of the speed profile of running vehicles can be used to establish the emission models based on the adjusted average speed. In practice, with the development of lighter, lower-drag design technologies that increase vehicle speed, it has been shown to significantly increase fuel efficiency and reduce emissions (Correia et al., 2014).

Instantaneous speed is a key indicator used by many instantaneous emission models. Instantaneous emission models usually relate the vehicle's instantaneous speed to its fuel consumption and/or emissions over a run cycle based on different parameter requirements (Boulter et al., 2007). It is used to map the emission test results of the chassis dynamometer or engine testbed under neutral conditions (Ajtay et al., 2005). The application of instantaneous emission models requires a higher accuracy of the vehicle's operation and position measurements.

The widely-used VSP binned emission factor method was applied to a multi-scale emission modeling method called MOVES, from project-level scale analysis to national inventory prediction (Sun et al., 2015; Ubiergo and Jin, 2016). The physical meaning of this method is the ratio of instantaneous power to the mass of the source (Jimenez-Palacios, 1998). The advantage of using VSP as the parameter of the operating state lies in that the instantaneous emissions of vehicles present a great regular change. The VSP is then based on a second-by-second measurement using vehicle speed and other parameters.

3.2. Road environmental factors

3.2.1. Segment factors

Vehicular operating factors are considered when the vehicle directly emits pollutants. Some scholars have realized that external environmental factors of vehicle operation need to be taken into account. Costagliola et al. (2018) proposed that the actual road conditions of vehicles during the operation had a significant impact on the emission results. Wood and other scholars noted that the actual road conditions of the vehicle would affect the engine power and braking system of the vehicle, thus affecting the energy consumption and pollution emissions of the vehicle. As the influence of road conditions on the vehicle was ignored in the laboratory test, the calculated value was smaller than the actual result (Della Ragione and Giovanni, 2016; Gallus et al., 2017; Prati et al., 2014, 2015; Wang et al., 2015; Wood et al., 2014). Frey et al. (2008) found that the neglect of the road conditions would result in a 16%–22% uncertainty in the evaluation results of pollution emissions. The consideration of the factors affecting the road conditions during vehicle operation is very important in the establishment of a vehicle pollution emission model.

The road environmental factors reflect the external environment dependent on the vehicle emissions in regards to segments and intersections. Segment factors include the road elevation, road grade, and the relationship between the tires and road surface. The elevation of the road has an impact on

the engine and fuel efficiency of vehicles: the higher the altitude, the greater the impact (Franco et al., 2013). Similarly, the change of the road's slope inevitably leads to the change of the vehicle's load. Generally speaking, a positive slope increases the load while a negative slope reduces the load, which eventually leads to the change of the vehicle pollution emission results (Della Ragione and Giovanni, 2016; Prati et al., 2015; Wang et al., 2015). Studies have shown that changes in the parameters between the tires and road surface can also stimulate the running efficiency of vehicles, thus increasing the impact of vehicle emissions (Hernandez et al., 2017; Kang et al., 2014; Kim et al., 2017; Wang et al., 2012a, 2014; Yang and Al-Qadi, 2017).

3.2.2. Intersection factors

In the urban road network, vehicle deceleration, idling, and acceleration frequently near intersections cause vehicle delays and produce greater energy consumption and emissions. The intersection factors have a significant impact on the emissions of vehicles running at intersections. These factors can be divided into the signal timing for isolated intersections, signal coordination for signalized arterials, and the average delay. For isolated intersections, the signal timing design directly influences the state of the vehicle's movement, including the accelerating, reinforcing, cruising, and idling of the vehicle. In other words, when vehicles are running at intersections, they are controlled by signals and adjust between different operating states, which inevitably affects the engine's operating efficiency. Thus, the vehicle emissions will change accordingly. The contributing trends of the vehicle acceleration, deceleration, and cruising to emissions are not the same. As the delay time of vehicles at the intersection becomes longer, the cumulative time of idling is also continuously extended, leading to the increase of the total emissions of the vehicles at the intersection.

The intersection signal designs have different impacts on vehicle emissions. When the scope of the study is a single arterial control system, each intersection is integrated with the signal joint control system. Emission analysis should not only consider the pollution emissions at a single intersection, but also analyze the pollution emissions at each intersection in the joint control system. The size of the pollution emissions is different from the sum of the emissions at a single intersection. In other words, the indicators of the signal coordination control also have an impact on the vehicle emissions in the coordinated control system.

4. Emission evaluation for operating vehicles

4.1. Fuel-based carbon emission analysis

This calculation method of the emissions based on the energy consumption provided by the IPCC determines the emissions using the given energy consumption data and energy carbon emission factor. The calculation method has a strong applicability and determines the carbon emission by combining the carbon emission factor of various types of energy in different industries. Therefore, this algorithm has been widely used in

research. Many studies have used the method in traffic emission calculations (Lyu et al., 2019; Pallavidino et al., 2014; Puliafito et al., 2015; Wang et al., 2012a).

This method is mainly applied to the measurement and analysis of transportation carbon emissions in cities, regions, and whole countries. The emission characteristics of different energy consumptions and different years are taken into consideration. For example, Mensink et al. (2000) established the urban transportation carbon emission model through the “bottom-up” method in the early stage and took the Antwerp region as a research object to measure the transportation carbon emissions. Later scholars (Creutzig and He, 2009; Liu et al., 2015) established the carbon emission model of urban transportation by the “top-down” or “bottom-up” methods, which mainly include variables such as the vehicle volume, vehicle mileage, vehicle fuel consumption, and energy carbon emission factor. Other studies used two methods to measure the traffic carbon emissions of regional roads (Saija and Romano, 2002; Zuo et al., 2018). Other researchers used two methods or combined them with the method of “top-down” or “bottom-up”, based on the different types of vehicles, ownership, mileage, unit mileage fuel consumption, and other data to calculate transportation energy consumption, so as to measure the national and regional scale transportation emissions (Han et al., 2018; Ramachandra and Shwetmala, 2009). Some national transport carbon emissions studies covering a span of several years show that the carbon emissions from China's vehicles are growing rapidly (Xie et al., 2016).

Based on the carbon emission analysis of the energy consumption, another research approach is to calculate and analyze the carbon emissions of different types of transportation models, including passenger and freight transportation (Creutzig and He, 2009; Liu et al., 2015; Palencia et al., 2017; Wang et al., 2012b), private automobiles (Creutzig and He, 2009), and public transportation (Zhang et al., 2012). Taking Beijing as an example, the “bottom-up” method was used to calculate the total carbon emissions from the transportation industry, and it was found that freight transportation and passenger cars were the main sources of the increase in carbon emissions (Creutzig and He, 2009).

4.2. Speed emission model application

Another important research direction is to estimate the total emissions of urban sections or regions by applying emission models based on the vehicle speed and related parameters, so a large number of emission models have been proposed and used.

Part of one study analyzed the vehicle trajectory velocity data of different driving conditions and different vehicle fleets from the perspective of the model's application and analyzed vehicle emissions by various emission models. By evaluating the emissions of different vehicle types, the IVE and MOBILE 6 models showed that the emission factor of trucks is much higher than that of light vehicles. In Beijing, the emission factor can be as high as 44 times that of light vehicles (Yao et al., 2006).

The results of a study on bus emissions based on the MOVES model showed that when a bus is in congested traffic, the emissions of the bus can be reduced by 5% for each additional

passenger compared to passenger cars (Alam and Hatzopoulou, 2014). Some scholars analyzed the taxi track data of Shanghai. Based on the average velocity of the track combined with the COPERT model, they analyzed the emissions of taxi areas and road networks in Shanghai (Luo et al., 2017). Some studies localized the emission rate in the MOVES model through the measured data and obtained a localized model conforming to the emission characteristics of the motor vehicles. The prediction effect of the model on the emission of cars, buses, and container vehicles was improved by 47%, 53%, and 70%, respectively (Cao, 2018).

In addition, the environmental scenarios have an impact on vehicle emissions due to different vehicle trajectories at different altitudes and in different seasons. After studying the development of vehicle emission factor measurement methods, Franco et al. (2013) believed that the actual temperature and altitude of the vehicle in operation had a certain influence on vehicle emissions. The IVE model was used to calculate the emissions of light commercial gasoline vehicles in the city, and the results showed that the emission rates of CO and VOCs were proportional to the altitude (Nagpure et al., 2011). Wallace et al. (2012) used the MOBILE model and the MOVES model to calculate the CO and emissions from vehicles on urban highways in winter. The results calculated by the MOVES model were closer to the actual observed values.

In recent years, in order to improve the accuracy of vehicle emission measurement, many scholars have combined emission models with some other methods and means to improve and modify the accuracy of the assessment results. GPS is a commonly used method to obtain a large number of vehicle driving tracks; the vehicle velocity and acceleration trajectory data are used as the input parameters, combined with vehicle emission models, such as MOVES model (Liu et al., 2013) and COPERT model (Tang et al., 2017), to estimate the emissions of vehicles or taxis on the road network and improve the accuracy of the vehicle emission estimation (Nyhan et al., 2016).

For the data collection of the actual vehicle operation mode and driving conditions, some scholars used the floating vehicle method or remote sensor to collect vehicle running track data. Using this information, the basic emission factor in the emission model can be corrected for regional differences. Combined with the revised emission model, the comprehensive vehicle emission factor (Dong et al., 2011; Shan et al., 2017) in line with the actual driving conditions in the study area was determined. Many other scholars used on-board emission testing systems, such as PEMS, to obtain instantaneous emissions during vehicle operation (Gallus et al., 2017; Yu et al., 2021).

In addition, in some studies, traffic simulation software, such as VISSIM and TRANSIMS, was integrated to obtain sufficient data of the average and instantaneous vehicle operation speed to accurately estimate exhaust emissions under the vehicle operation mode (Abou-Senna et al., 2013).

In the past few years, some scholars have proposed new vehicle emission factor calculation models and verified the estimation accuracy of the proposed new methods by making detailed comparisons with the estimation results of existing emission models. For example, some scholars proposed a new

approach to estimate automobile emissions and estimated the total fuel consumption and emissions of the Madrid Municipal Transport Company in one year. By comparing these data with the actual results, they showed that the estimation error of the proposed method was acceptable (López-Martínez et al., 2017). Han et al. (2019) proposed a new regional vehicle emission factor calculation model, which was modified without considering the environmental parameters. By calculating the daily average emission factor of small vehicles, the model was compared in detail with the calculation results of COPERT-V model, and the experimental results showed that the model achieved good results.

4.3. Emission analysis for vehicles operating at intersections

To improve the urban traffic environment, we should focus on reducing vehicle emissions at intersections when measuring vehicle operating emissions.

Over the past few decades, the primary focus of vehicle emission models was to develop a reliable mathematical correlation between the characteristics of vehicles running at intersections and their emissions, which could help policy-makers and traffic engineers in planning and developing more significant and effective countermeasures to control the emissions in transportation. Various emission models have been developed and applied for analyzing the relationship of the traffic speed, speed change, signal cooperation, and intersection vehicle emissions.

4.3.1. Intersection vehicle emission analysis

A basic algorithm for intersection emission analysis that converts CO emission factors from travel speed models to the modal CO emission rate is employed by the US Federal Highway Administrative model CALINE 4 (Benson, 1984).

Liao and Machemehl (1998) analyzed the speed changes of vehicles at different signal stages when they are running on the entrance, intersection, and exit roads of the intersection. An intersection aggregate energy consumption model (AFCM) was proposed, and based on this model, the influence of the signal timing on the vehicle energy consumption was studied. Researchers further considered the influence of random vehicle driving on fuel consumption, and deduced and solved the optimal signal period (Liao, 2013) with the minimum fuel consumption.

In 2000, the impact of traffic signals on emissions using the mobile emissions assessment system for urban and regional evaluation (MEASURE) model considering the signal timing factor on the CO emissions was investigated by Hallmark et al. (2000), who found a notable decrease in the CO emissions when the delay time at the traffic signals was reduced. In 2001, the impact of the traffic flow on real-time vehicle emissions was investigated by Roupail's group (Roupail et al., 2001). The vehicle emissions rates were evaluated when the vehicle was accelerating, decelerating, cruising, and idling. The relationship between the control delay and vehicle emissions was analyzed. They discovered that the vehicle emissions tended to be higher during the transition

from stop to acceleration and the emissions were relatively lower during idling. The phenomenon that the quantity of vehicle emissions was approximately double in the control delay compared with cruising was also observed.

Garber and Hoel (2014) employed the classic theory of Weber and Cobbe for signal timing optimization of an isolated intersection to minimize the delay time and found that the fuel consumption of vehicles is high at signalized junctions.

Midenet et al. (2004) used a multi-camera system for testing vehicle idle time and stop rates. These data, which were affected by each vehicle whether it was stopped or idling at a red light, were employed to quantify CO₂ emissions, fuel consumption, and emission cost for an isolated signalized intersection.

Coelho et al. (2005) applied experimental measurements to interpret the connections between the traffic emissions, drivers' behavior, traffic variables, and signal control settings in terms of the minimum signal settings, minimum green time setting, and speed threshold setting. The discovery that signal control schemes designed to prevent speeding resulted in higher emissions was one of the conclusions. However, if drivers' behavior changed by reducing speed by adjusting the traffic signals for speed control, a reduction of pollutant emissions resulted. The conclusion that the emissions of CO, NO, and HC increased up to 15%, 10%, and 40%, respectively, and the signal presence was obtained based on the data gathered in Portugal on Highway N6, which links the cities of Lisbon and Cascais. One important finding of their research was that it was not suitable to use macroscopic models employing fixed driving cycles (such as the EMFAC, COPERT, and MOBILE to analyze vehicle emission changes caused by traffic signal adjustments for speed control.

Pandian et al. (2009) found the relationships of the intersection traffic and vehicle characteristics and vehicle emissions were explained by evaluating the most likely characteristics influencing vehicle emissions at the intersection based on the summarization of the traffic flow and emission models.

To evaluate the effects of signal timing on vehicle emissions at a pre-timed signalized intersection, Wu and Li et al. (2011) employed a two-stage approach. The first stage was used to measure the trade-off between the vehicle delays and the number of stops by applying a dynamic programming algorithm for optimization models; the second stage used microscopic emission estimation models integrated with vehicle-specific power methods to estimate vehicle emissions on the basis of the results from the first stage.

Huang et al. (2014) analyzed the running characteristics of vehicles at signalized intersections and summarized the cumulative fuel consumption model of vehicles at signalized intersections by analyzing vehicle deceleration, idle speed, and accelerating fuel consumption.

Additional research (Zhang et al., 2014) resulted in a cumulative fuel consumption model at intersections based on the premise of reducing the vehicle fuel consumption and delays at intersections, and combined it with the delay

model to realize signal timing optimization by using the particle swarm optimization algorithm.

To solve the problem of the high fuel consumption of vehicles at over-saturated signalized intersections, Zhao et al. (2017) took the duration of the entire over-saturated traffic state at signalized intersections as the research period, analyzed the queue length, stops, and passing time of vehicles by using the deterministic theory, and determined the deceleration, idling, acceleration, and uniform running time of vehicles at signalized intersections. Furthermore, using the fuel consumption rate of vehicles under different driving cycles, the average fuel consumption model of all vehicles from the first stop to crossing the stop line at the over-saturated intersection was established.

These studies on the analysis of the vehicle exhaust emissions when passing signalized intersections were based on actual data collected and some specific assumptions. Moreover, this kind of analysis is based on the relevant traffic parameters of a single intersection and studies the exhaust emissions of vehicles operating at the intersection. Other studies have a perspective on coordinated intersection systems at arterials to analyze vehicle emissions and characteristics as vehicles move through the system.

In 2000, Rakha's group (Rakha et al., 2000) proposed that emissions can be decreased up to 50% by productive signal coordination. Unal et al. (2003) proposed a model to examine the impact of traffic signal timing and coordination on vehicle emissions at signalized arterials on the basis of field data for an experimental vehicle emission test, which was conducted for four drivers and eight light-duty gasoline vehicles on Walnut Street and Chapel Hill Road in Cary, North Carolina. This real-world vehicle emission measurement tested 100 h of 824 one-way runs and 2020 vehicle miles traveled on two signalized arterials; it was concluded that acceleration was correlated with the highest emissions, and cruising, deceleration, and idling were related with lower emissions, in descending order. Zhang et al. (2009) employed the portable emission measurement system (PEMS) to gather the field data of vehicle emissions to compare the emission levels and analyze emission distributions in Beijing with two signal control scenarios: signal coordination and signal non-coordination. A simulation platform integrated with VISSIM and the vehicle specific power (VSP) was established in the study to assess the microscopic traffic emissions affected by the signal coordination, alternative signal timing, and traffic flow.

Moreover, the impact of traffic signal coordination on emissions and operational performance measures of delay and stops was studied by Lyu and Zhang (2012). Results were obtained by simulating cycle lengths and offsets and comparing different platoon ratios. First, it was observed that the effects of the cycle length on delay were greater than those on stops and emissions for undersaturated intersections. Second, the delay, stops, and emissions all decreased with the platoon ratio increase if the cycle length was a fixed value, indicating that the emission reduction was more significantly associated with stops than delays. Moreover, given a fixed cycle length and platoon ratio, the

early platoon arrival was more likely to correlate with higher emissions than the late platoon arrival.

4.3.2. Intersection vehicle emission analysis via simulation

Due to a large number of experimental sample data that often need to be collected in the study of vehicle emissions and the requirements for research conditions, model applicability is limited to a certain extent. To solve this problem, some scholars combined traffic simulation software and vehicle fuel consumption and emission models to analyze vehicle emissions at signalized intersections.

Zhang et al. (2009) built a micro-traffic exhaust simulation platform by combining VISSIM with the VSP modeling method, and evaluated the exhaust emissions of signalized intersections under two traffic control strategies of different signal timings and different traffic flows. Liu et al. (2010) combined VISSIM and the micro-energy consumption emission model to build a micro-model platform that can simulate the relationship between vehicle instantaneous running conditions and emissions, and quantitatively analyzed the influence of signal cycle change at signalized intersections on vehicle exhaust emissions. Lyu and Zhang (2012) studied the influence of the coordinated control of adjacent intersections on vehicle exhaust emissions by combining the VISSIM and MOVES models. Gao and Hu (2015) performed a simulation of the traffic flow characteristics at signal-controlled intersections using VISSIM to obtain real-time vehicle data, including the location, time, vehicle type, and vehicle speed. This data was imported into the VERSIT + model to obtain the total exhaust emissions of queuing vehicles at the intersection.

In addition, other researchers (Kwak et al., 2012) attempted to optimize traffic signal timing by evaluating the effects of signal timing optimization on vehicle fuel consumption and emissions in an urban corridor and maximizing the environmental benefits by combining the genetic algorithm, transportation analysis simulation system (TRANSIMS), and VT-Micro emission model.

Through the combination of traffic simulation software, such as VISSIM, and different fuel consumption emission models, researchers analyzed vehicle emissions at signalized intersections to overcome the limitations of the experimental conditions. However, the combination of the VISSIM and emission model resulted in a complex process and a large amount of data processing capacity. As a result, there are still some deficiencies in the research on the emissions generated by the traffic flow at signalized intersections based on the combination of VISSIM and the vehicle emission model.

In recent years, some scholars have combined the car following model with the emission model to analyze the influence of the vehicle trajectory and signal design on vehicle emissions, and they obtained some reasonable conclusions. For instance, Tang et al. (2017) analyzed the influence of the green signal ratio on vehicle fuel consumption and emissions by adopting the car following model to simulate the vehicle driving trajectory and combined this with the VT-Micro model. He et al. (2018) studied the influence of vehicle lane change on fuel consumption of the traffic flow by the car following model. Their research results showed that a

reasonable lane changing scheme had a positive impact on reducing the fuel consumption of the traffic flow.

5. Conclusions

Through combining the existing research results on operating vehicle emissions, it was found that vehicle pollution has become an important source of air pollution with the accelerated development of urbanization. As traffic emissions are prominent contributors of greenhouse gases, they are critical areas of study in the field of traffic research.

At present, although there is a large number of studies for the in-depth analysis of vehicle emissions at segments and intersections, there is little in the literature on operating vehicle emissions with newer transportation technology. The literature on operating vehicle emissions can be summarized as follows.

- (1) From the perspective of the research scope, studies mostly focused on the analysis of vehicle emissions at the micro level to evaluate the emissions of certain vehicle types or the study of vehicle emissions in a certain area from the meso-level of the road network. Moreover, the estimation of vehicle emissions for isolated intersections or signalized streets along corridors using microscopic modular or averaged-speed-based emission models may result in a reasonable estimate of the expected average vehicle emissions. Some proposed models are only applicable when both intersection data and segment data are available for the specific roadway network conditions for which the estimate is being made. However, the vehicle emission analysis could be improved by combining related intersections and segments and extending the work to a dense area within an urban street network. While several system-wide analysis methods exist that can compensate for the potential lack in connectedness and coordination control resulting from independent site analysis, it is still necessary to research network emission models with both stopped vehicles at intersections and cruising vehicles at segments using dynamic traffic parameters affected by the network to combine the two estimates into an average.
- (2) From the perspective of the research objects, the current research on vehicle emissions often focuses on traditional fuel vehicles. However, as new energy vehicles become the direction of future automobile development, their energy consumption and emission characteristics in different road environments need to be further analyzed. The measurements of the energy consumption and pollution emissions of new energy vehicles have mainly used data from laboratory tests. A quantitative mathematical model system of the energy consumption and pollution emissions (Kelly and Rajagopalan, 2001; Nam et al., 2005) was established through the simulation of the running trajectory of selected vehicles.

Some scholars have carried out detailed research on the vehicle running data and vehicle types simulated in the laboratory, and developed calculation models (Zanni and Bristow, 2010) based on the correlation between the emissions and driving distance, driving frequency, fuel consumption in kilometers, physical properties of fuel, and corresponding emission factors. Frey et al. (2006) established a path-based emission factor measurement model by combining the detection model of the EPA to compare the emission factors of new-fuel vehicles and conventional vehicles. After the development of the MOVES model by the EPA, some scholars established the emission model (Wu et al., 2015a; Zhai et al., 2011) of HEV on the theoretical basis of VSP, taking into account the start and stop mechanisms of the internal combustion engine, engine type, average driving speed, and other factors. Some studies have evaluated the energy consumption of electric vehicles. By analyzing the force of vehicles in the process of driving, an estimation model of the same type of electric vehicles' instantaneous energy consumption was proposed from the perspective of vehicle energy conversion, and the prediction accuracy was guaranteed to a certain extent (Abouleiman and Rawashdeh, 2015; De Cauwer et al., 2015; Genikomsakis and Mitrentsis, 2017; Wu et al., 2015b).

Although these studies have played a role in understanding the mechanism and quantification of the estimation of energy consumption and emissions of new energy vehicles, the total energy consumption and emissions of new energy vehicles running on different segments and intersections deserve further study. The development of intelligent transportation systems and autonomous driving vehicle technology means that the cruise control of modern driving vehicles is fundamentally different from that of traditional manual driving, and the analysis of the impact of cruise control on vehicle operating emissions in the environment of the connected vehicles needs to be addressed in future studies.

- (3) Most of the research achievements on operating vehicle emissions focus on the application of existing emission models or the revision and localization of existing emission models. The control of a vehicle's running state is mainly based on laboratory data, field data, or simulation results. However, with the rapid development of new energy vehicle technology and the improvement of intelligent transportation systems, the existing vehicle operating emission model method needs to be updated to meet the emission assessment needs of these new technology vehicles. Laboratory data collection is based on certain assumptions and field data collection is a complex process that involves a large data processing capacity, which limit the applicability of the research results. In addition, the actual road emission models of vehicles only focus on the study of characteristics of operating vehicles and do not consider the vehicle's energy consumption and emission characteristics under complex road conditions, such as the impact of road type differences on the vehicle's operating energy efficiency and emissions. Therefore, there is a need for additional research on

new technology vehicles for the quantitative analysis of the operating characteristics and emissions.

Conflict of interest

The authors do not have any conflict of interest with other entities or researchers.

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